PRODUCT REQUIREMENT DOCUMENT (PRD)

SmartRent AI : Decision Intelligente Engine

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Project Name: SmartRent AI (Decision Intelligente Engine)

Version: 13.0

1. Project Overview

1.1 Problem Statement:

Many renters make poor housing choices because they focus only on the monthly rent price. These individuals suffer from "economic blindness" when they fail to account for fixed financial commitments, commute distances, and high upfront deposits, which often lead to long-term financial instability.

1.2 Target Domain:

The target domain for this project is the rapid visualization of urban housing and budget optimization.

1.3 Proposed Solution Summary:

SmartRent AI removes the hassle of manual budgeting and research by allowing users to input their salary, commitments, and transport data alongside various rental advertisements. This system generates a high-fidelity economic mock-up instantly, effectively shifting the user's role from a manual researcher to a Financial Curator.

2. Background & Business Objective

2.1 Background of the Problem:

For many years, the process of evaluating rentals and personal financial health followed a manual, linear pathway where users had to guess their leftover cash after paying bills and travel expenses. This traditional method requires extensive labour and financial awareness skills that many people lack, effectively blocking them from visualizing the true "Real Cost of Living".

2.2 Importance of Solving This Issue:

Accelerating the preparation of an economic mock-up from hours to seconds allows users to test multiple housing scenarios including commute distances and deposit requirements and discard unsustainable options immediately. Skill barriers are broken down so that any user can visualize their financial intent clearly without the need for years of training in economics.

2.3 Strategic Fit / Impact:

The project aligns perfectly with the goal of Economic Empowerment using Z AI GLM to handle complex financial reasoning. Fresh graduates and urban workers can protect their disposable income and plan for a better future using this technology.

3. Product Purpose

3.1 Main Goal of the System:

The main goal of the system is to provide an AI partner that optimizes housing decisions and overall financial health.

3.2 Intended Users (Target Audience):

Intended users for this platform include fresh graduates entering the workforce, new workers moving to urban centres, and budget-conscious entrepreneurs.

4. System Functionalities

4.1 Description:

The system operates as a high-speed generative engine for financial design and budgeting. Using a Prompt-to-Comparison model, the AI captures the complex relationships between Salary, Fixed Commitments, Commute Distance, and Rental Ads to render a clear multi-screen comparison layout.

4.2 Key Functionalities:

- Economic Vibe Designing: This feature translates raw, messy rental ads and personal financial data into high-fidelity budget designs.

- Conversational Logic Editing: Users can interact directly with the results canvas using natural language to update variables like transport mode or budget instantly.
- Financial Versioning & Infinite Canvas: AI-enabled tracking allows users to compare dozens of different housing scenarios on a single 2D workspace.
- Export - Component Based: The system converts final comparisons into code snippets or summary reports for easy record-keeping.

4.3 AI Model & Prompt Design

4.3.1 Model Selection:

The system uses Z AI GLM as the primary large language model. This model is the right choice because it has strong capabilities for multi-step reasoning and generating structured output from unstructured text.

4.3.2 Prompting Strategy:

Our team uses a multi-step agentic prompting approach to process rental advertisements. This strategy fits our use case because it ensures the AI follows a clear logical path when calculating the real cost of living for the user.

4.3.3 Context & Input Handling:

The system accepts a maximum input size of 1500 tokens before it starts to cut or reject data. Any inputs that exceed this limit will undergo truncation or chunking to keep the analysis efficient and within the AI's memory limits.

4.3.4 Fallback & Failure Behaviour:

A graceful error state is triggered if the model returns an off-topic or hallucinated response. The product includes a mechanism for retrying and a human check path so that users can verify the information and reach a correct decision.

5. User Stories & Use Cases

5.1 User Stories:

- As a fresh graduate starting a first job in the city, I want to enter my salary and fixed monthly commitments so that I can find a room that does not exceed my realistic budget.
- As an urban commuter, I want to input my daily commute distance and preferred transportation mode so that I can see the total cost of travel before I sign a rental agreement.
- As a smart renter looking for the best deal, I want to paste multiple room advertisements so that the AI can rank them based on my actual remaining cash at the end of every month.

5.2 Use Cases (Main Interactions):

5.2.1 Prompt to Analysis Flow:

- End User enters their Financial Profile, which includes monthly salary, fixed commitments, and maximum deposit budget.
- End User provides specific Commute Logistics such as workplace (area), transport method (car/motor, public transport and walk/bicycle) and maximum distance preferred.
- End User pastes Rental Ad Data (text or link) into the SmartRent AI for processing.
- SmartRent AI bundles these inputs and triggers the Z.AI GLM API to parse the rental price and key facilities.
- The system renders the Comparison Canvas to display the total "Real Cost" and an initial economic ranking.

5.2.2 Multi-Option Ranking Flow:

- End User triggers the "Add Another" section in the SmartRent AI to compare up to five different sets of Rental Ad Data.
- Z.AI GLM API processes all options simultaneously while maintaining the constraints of the Financial Profile.
- The system generates a comprehensive ranking on the Comparison Canvas, sorting choices from the most affordable to the most expensive.
- End User interacts with the Comparison Canvas by typing natural language commands into the integrated chat interface to update their Commute Logistics or Financial Profile.

- Z.AI GLM API identifies the updated intent and instantly reflects the new "Real Cost" within the visual ranking boxes.
- The Z.AI GLM engine performs advanced **AI Intent Parsing** to accurately identify natural language commands from the user (e.g., 'Update AD 1 rent to 900'). This allows the system to instantly recalculate the 'Real Cost' and refresh the visual ranking on the Comparison Canvas without requiring a full form resubmission.

5.2.3 Use Case Diagram:

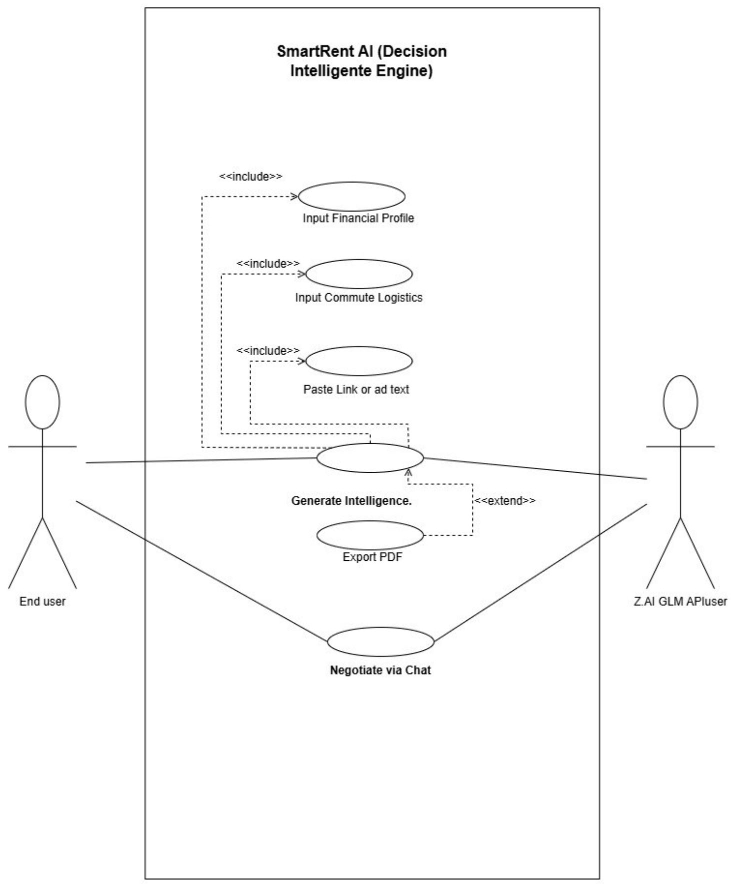

Figure 1: Use case Diagram

6. Features Included (Scope Definition)

- **Interactive 2D Comparison Canvas:** A dynamic visual workspace where users can generate, view, and refine housing rankings in real-time.
- **Unstructured Data Parsing:** Advanced processing of messy rental advertisements via text-to-analysis and link-to-analysis inputs, handled by the Z.AI GLM engine.

- **Specialized Financial & Logistics Modules:** Dedicated input fields to capture critical user data, including monthly salary, fixed financial commitments, and commute logistics (distance and transport mode).
- **AI-Driven "Real Cost" Reasoning:** Integration of the Z.AI GLM as the primary reasoning engine to perform multi-step calculations, uncovering the true cost of living for every rental option.
- **Standardized Sustainability Safety Buffer:** Implementation of a mandatory RM 550 safety buffer within the reasoning logic. The system automatically flags properties as "RISK" if the estimated disposable income falls below this threshold.
- **Conversational Logic Editing:** A chat-based interface integrated into the canvas that utilizes **AI Intent Parsing** to update financial variables and commute data instantly through natural language.
- **Consistent Economic Logic:** Application of a coherent mathematical framework across the entire system to ensure stable, reliable, and standardized budget calculations.
- **One-Click Report Export:** Functionality allowing users to export their final economic analysis and ranked comparison as a summary report or PDF for offline record-keeping.

7. Features Not Included (Scope Control)

- Implementation of backend logic, full APIs, and a permanent database system is not included in this development phase.
- This project serves only as a frontend scaffolding for the prototype and does not support a full production-ready environment.
- Simultaneous inputs from multiple users or live collaboration features are not supported in the initial version of the product.
- Real-time GPS location tracking and automated distance calculation using map APIs remain outside the current scope to keep the project focused on AI reasoning.

- Automated payment processing for rental deposits is excluded from the current version to ensure user data security during the hackathon.

8. Assumptions & Constraints

- LLM Cost Constraint: Token costs per average user session are estimated to be manageable through the use of local caching for repeated queries. Our team has decided to limit the number of active comparison sessions to keep inference costs under control.
- Technical Constraints: Code export features remain limited to basic HTML and CSS formats, as modern frameworks like React or Swift are not supported.
- Performance Constraints: High-quality visual generation requires many tokens, so the system includes a daily generation limit to maintain stable performance.
- User Input: The model is unable to generate meaningful results without clear human prompts and requires final human approval before any decision is finalized.

9. Risks & Questions Throughout Development

- Design Similarity: Our team is investigating how to ensure the AI does not generate identical housing layouts or budget designs for different users across the platform.
- Frame Stability: The user might need to manually fix the layout or numerical data if the system produces a broken interface after exporting.
- Data Interpretation: Issues may arise if the AI is unable to correctly contextualize the specific text or short descriptions provided in a rental advertisement.
- Calculation Accuracy: A potential risk exists where the AI might misinterpret local Malaysian currency formats if the advertisement text is too messy or unstructured.